

APOORVA KHAIRNAR

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EDUCATION

Ph.D. in Mechanical Engineering

Expected May 2028

Virginia Tech, Blacksburg, VA

GPA: 3.97/4.00

Research: Soft robotics, robot learning, reservoir computing, neuromorphic computing, sensing, control

M.S. in Mechanical Engineering

May 2026

Virginia Tech, Blacksburg, VA

GPA: 3.97/4.00

B.Tech. in Mechanical Engineering, Honors in Thermal Engineering

June 2021

CoEP Technological University, Pune, MH, India

GPA: 4.00/4.00

RESEARCH & WORK EXPERIENCE

Graduate Research Assistant - Department of Mechanical Engineering, Virginia Tech

Aug 2023–Present

- Built spider-web-inspired compliant fiber network testbeds for physical reservoir computing and soft robotic sensing
- Developed video-analysis pipelines to track fiber-network deformation from experimental recordings
- Quantified physical reservoir performance using memory capacity and nonlinearity metrics
- Developed optimization frameworks for spiking reservoirs using Nengo, Lava, Intel Loihi, and evolutionary algorithms
- Fabricated hydraulic artificial muscles achieving 15% contraction with 10 mL volume change for underwater soft robots

Software Developer - Siemens Technology and Services Pvt. Ltd., Pune, India

Aug 2021–Jun 2022

- Developed C++/MFC application for automation and configuration of HVAC control systems
- Implemented and debugged engineering software workflows involving user interfaces

Computational Fluid Dynamics Intern - Intelimek Systems, Pune (India)

Jun 2021–Aug 2021

- Contributed to digital-twin development for fermentation and fluid-flow simulations in pharmaceutical applications
- Supported workflows integrating machine learning models with CFD simulation data

Summer Research Intern - Indian Institute of Technology, Kanpur (India)

Aug 2020–Oct 2020

- Built a transient axisymmetric nozzle-flow model in ANSYS Fluent
- Visualized downstream vortex formation using time-resolved vorticity contours

PROJECTS

Model Predictive Control for Unmanned Ground Vehicle Trajectory Tracking

- Implemented MPC-based trajectory tracking for an unmanned ground vehicle using a dynamic bicycle model

Trajectory Tracking of 4-DOF Kinova Arm

- Implemented and compared Lyapunov, adaptive Lyapunov, and impedance controllers for simulated 4-DOF Kinova arm

Kirigami Vacuum Jamming Gripper

- Designed and prototyped a kirigami-inspired vacuum-jamming soft robotic end-effector

PUBLICATIONS

- Naughton, Tekinalp, Shivam, Kim, **Apoorva Khairnar**, Kindratenko, and Gazzola, "[Neural reservoir control of a bio-hybrid soft arm](#)," *Proceedings of the National Academy of Sciences*, 2026
- **Apoorva Khairnar**, Phalak, Wang, Zhou, Rümmelein, Jantzen, Li, and Naughton, "[Physical reservoir computing with compliant fiber networks](#)," in *ASME Smart Materials, Adaptive Structures and Intelligent Systems*, 2025
- Sivashangaran, **Apoorva Khairnar**, and Eskandarian, "[Mobile Robot Exploration Without Maps via Out-of-Distribution Deep Reinforcement Learning](#)," *IFAC-PapersOnLine*, 2025
- **Apoorva Khairnar**, Sivashangaran, and Eskandarian, "[A Comparison of Motion Planning Methods for Autonomous Ground Vehicle Exploration and Search](#)," in *ASME International Mechanical Engineering Congress and Exposition*, 2023

TECHNICAL SKILLS

Robotics & Control: Soft robotics, robot learning, reinforcement learning, learning-based control, motion planning, soft robotic sensing

Machine Learning: Neural reservoir computing, physical reservoir computing, spiking neural networks, optimization, evolutionary algorithms

Programming & Tools: Python, MATLAB, C++ (MFC), Nengo, Intel Loihi, Lava-NC, Stable-Baselines3, Git, Linux, Gymnasium, PyElastica, OpenCV, PyBullet

Experimental Systems: Soft actuator fabrication, compliant structures, sensing/data pipelines, deformation tracking, hardware testing

AWARDS, PATENTS & SERVICE

- Co-inventor, provisional patent application in preparation: Physical reservoir fiber network sensor **Apr 2026**
- Peer Reviewer: ISA Transactions, ASME IMECE, MECC-JAVS, IEEE-TVT **2024–Present**
- 1st Place, PhD Research Flash Talk, GPSS Symposium, Virginia Tech **Mar 2025**
- Duly Elected Full Member, Sigma Xi Scientific Research Honor Society **Apr 2025**
- Led robotics outreach by fabricating 10+ silicone soft arms and presenting demos to 100+ participants **2025–2026**